Unit 4 ENZYMES



Anselme Payen

Anselme Payen was a french chemist who first isolated the enzyme diastase in 1833 and showed that it can catalyze the conversion of starch to maltose. However only in 1926 enzymes were shown to be purely made of proteins by James B Sumner, who crystallized the enzyme urease to confirm its purity.

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Learning Objectives

After studying this unit the students will be able to:

- Classify the enzymes based on their function and assign EC numbers.
- Describe the factors influencing the enzyme activity.
- Explain the different types of enzyme inhibitors.
- Explain isoenzymes.
- List the uses of enzymes in various fields such as medicine, industries, scientific research etc...

Introduction

Enzymes are a type of protein present in all living things. Enzymes are produced in animal, bacteria, fungi, yeast and plants. Enzymes act as biological catalysts, increasing the rate of chemical reactions without undergoing any permanent change to themselves. Enzymes are specific in their catalytic activity. They are colloidal in nature and thermo-labile in character.

Enzymes are used to speed up the biochemical reactions by nature. The reactions can be a transformation of chemical compound or a protein. For example, in our stomach, enzymes

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